

ADITYA KUMAR SINGH

AI & DS Engineering Student | ML & Algorithmic Developer | Competitive Programmer
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PROFESSIONAL SUMMARY

Driven AI & Data Science undergraduate with a strong foundation in algorithmic problem-solving and practical machine learning experience. Proven ability to architect **real-time facial recognition systems** on edge hardware and develop **data-driven scripting extensions**. Passionate about applying **mathematical expertise** to solve complex ML engineering challenges.

EDUCATION

Madhav Institute of Technology & Science | Gwalior 2024 - 2028
Bachelor of Technology (B.Tech) – Artificial Intelligence & Data Science CGPA : 7.88

- **Relevant Coursework:** Data Science (Supervised & Unsupervised Learning), Probability & Statistics, Linear Algebra, Multivariable Calculus (Area, Volume, Partial Derivatives), Theory of Computation (NFA/DFA conversion, Mealy/Moore tables), Software Engineering (V-Model).
- **Academic Achievements:** Ranked in the Top 20 Competitive Programmers college-wide; successfully completed over 700 LeetCode challenges using C++

WORK EXPERIENCE

Lead Developer | Part-time May 2024 - Present
Optimus Remote • Pune

- Led architecture of quantitative tools with dynamic risk-to-reward parameters, improving backtesting execution efficiency by **35%**.
- Optimized high-accuracy pipelines processing **10+ years of historical OHLC data** to refine high-probability swing trading setups.
- Implemented advanced market structure logic into automated screeners, enhancing risk management and saving **15+ manual hours** weekly.

Technologies / Skills Used : Pine Script, Python, JavaScript, SQL, REST APIs, Git, Docker

Algorithmic Strategy & Web Developer | Freelance Jun 2021 - Mar 2024
Independent

- Engineered and delivered **300+ custom automation projects**, maintaining a **98% client satisfaction rate** across execution bots and market screeners.
- Developed full-stack dashboards using the MERN stack, **reducing live metric tracking latency by 40%** for active users.
- Programmed advanced rule-based logic and fractal models using **Pine Script** and **Python** to identify complex market behaviors.

Technologies / Skills Used : Node.js (MERN), Pine Script, Python, REST APIs, Git

PROJECTS

Real-Time Face Recognition & Logging System (Edge AI) | 🐙 Github

- Deployed a high-throughput computer vision pipeline on **NVIDIA Jetson Nano** utilizing **NVIDIA DeepStream**, achieving **30+ FPS** real-time video processing.
- Optimized an **ArcFace** facial recognition model via **ONNX** conversion, **reducing inference latency by 40%** and enabling rapid edge data logging.

Technologies / Tools Used : Python, C++, DeepStream, ONNX, ArcFace, Jetson Nano

MITs Academic Result Extractor | 🐙 Github △ Vercel

- Developed an automated data extraction tool to scrape and parse university examination results, **saving 10+ hours** of manual data entry per semester.
- Streamlined the data collection process, successfully aggregating academic records for **over 500+ students** into structured formats.

Technologies / Tools Used : Python, FastAPI, Hugging Face TrOCR, ONNX, Selenium, Docker, Google Cloud APIs, SQLite

TradingView Strategy Optimizer (Chrome Extension) | 🐙 Github

- Engineered a **Manifest V3** extension to fully automate algorithmic strategy optimization, **accelerating backtest execution speeds by over 50x**.
- Utilized **Pine Script** to analyze fractal market structures, processing **10,000+ historical data points** to extract high-probability setups.

Technologies / Tools Used : JavaScript, Pine Script, HTML/CSS, Chrome Developer Tools

SKILLS

Languages :	C, C++, Python, JavaScript, TypeScript, Java, Pine Script
ML & Data Science :	DeepStream, ONNX, ArcFace, Hugging Face TrOCR, Supervised & Unsupervised Learning, Computer Vision
Web & Infrastructure :	Node.js, React, Next.js, Express, Flask, Docker, Kubernetes, Git, CI/CD, Linux
Databases :	MySQL, MongoDB, PostgreSQL, SQLite, Redis, Supabase
Core Fundamentals :	Data Structures & Algorithms (Top 20 College Rank, 700+ LeetCode), OOP, DBMS, Linear Algebra, Probability & Statistics